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Systematic review

Bone mineral density and fractures after surgical menopause: systematic review and meta-analysis

IE Fakkert,[a1†] N Teixeira,[a†] EM Abma,[b] RHJA Slart,[c,d] MJE Mourits,[e] GH de Bock[a]

[a] Department of Epidemiology, University Medical Center Groningen, University of Groningen, Groningen, the Netherlands [b] Division of Geriatric Medicine, Department of Internal Medicine, University Medical Center Groningen, University of Groningen, Groningen, the Netherlands [c] Department of Nuclear Medicine and Molecular Imaging, Medical Imaging Center, University Medical Center Groningen, University of Groningen, Groningen, the Netherlands [d] Department of Biomedical Photonic Imaging, University of Twente, Enschede, the Netherlands [e] Department of Gynaecology, University Medical Center Groningen, University of Groningen, Groningen, the Netherlands
Correspondence: IE Fakkert, Department of Epidemiology, University Medical Center Groningen, University of Groningen, PO Box 30.001, 9700 RB Groningen, the Netherlands. Email i.e.fakkert@umcg.nl
[1]Present address: Department of Human Genetics, Radboud University Medical Center, Nijmegen, PO box 9101, Nijmegen 6500 HB, the Netherlands

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Background Oophorectomy is recommended for women at increased risk for ovarian cancer. When performed at premenopausal age oophorectomy induces acute surgical menopause, with unwanted consequences.

Objective To investigate bone mineral density (BMD) and fracture prevalence after surgical menopause.

Search strategy A literature search of PubMed, EMBASE and Cochrane library was performed with no date restriction. Date of last search was March 1st, 2016.

Selection criteria Primary studies reporting on BMD, *T*-scores or fracture prevalence in women with surgical menopause and age-matched control groups.

Data collection and analysis Data were extracted on BMD (g/cm^2), *T*-scores and fracture prevalence in women with surgical menopause and control groups. Quality was assessed by an adaptation of the Downs and Black checklist. Random effects models were used to meta-analyse results of studies reporting on BMD or fracture rates.

Main results Seventeen studies were included, comprising 43 386 women with surgical menopause. Ten studies provided sufficient data for meta-analysis. BMD after surgical menopause was significantly lower than in premenopausal age-matched women [mean difference lumbar spine, −0.15 g/cm^2 (95% CI, −0.19 to −0.11 g/cm^2); femoral neck, −0.17 g/cm^2 (95% CI, −0.23 to −0.11 g/cm^2)] but not lower than in women with natural menopause [lumbar spine, −0.02 g/cm^2 (95% CI, −0.04 to 0.00 g/cm^2); femoral neck, 0.04 g/cm^2 (95% CI, −0.09 to 0.16 g/cm^2)]. Hip fracture rate was not higher after surgical menopause compared with natural menopause [hazard ratio: 0.85 (95% CI, 0.70 to 1.04)].

Author's conclusions No evident effect of surgical menopause was observed on BMD and fracture prevalence compared with natural menopause. However, available studies are prone to bias and need to be interpreted with caution.

Keywords Bone mineral density, fractures, oophorectomy, surgical menopause.

Tweetable abstract Bone health after menopause: no evidence for additional effect of surgical menopause on BMD and fractures.

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Introduction

Premenopausal oophorectomy may have several indications, such as benign or malignant ovarian tumours, or risk-reducing salpingo-oophorectomy (RRSO) in women at increased hereditary risk for ovarian cancer, or at the time of elective hysterectomy. Especially for RRSO, benefits and adverse consequences of surgical menopause as induced by premenopausal oophorectomy need to be weighed carefully. In the past, oophorectomy was recommended to many premenopausal women during hysterectomy, mostly to reduce ovarian cancer risk and prevent ovarian surgery in the future.[1] Nowadays, premenopausal RRSO is mainly advised for women at increased hereditary risk for ovarian

[†]These authors contributed equally to this work.

cancer, since there are no effective screening methods for ovarian cancer.[2] RRSO in these women reduces ovarian cancer risk by up to 96%, and mortality by up to 76%.[3–5]

Premenopausal RRSO induces surgical menopause, leading to an earlier and acute start of menopause. After natural menopause bone mineral density (BMD) decreases and fracture prevalence increases. It is unclear whether surgical menopause enhances this effect.[6] Many observational studies on BMD and fracture rate after surgical menopause have been conducted, examining women with therapeutic oophorectomy or RRSO. However, the results reported in these studies are inconsistent regarding the nature and severity of the effect of surgical menopause on bone health. American and British guidelines on osteoporosis recommend fracture risk assessment in all women after age 65, and earlier for women with untreated premature menopause.[7–10] British guidelines on familial breast cancer and on menopause advise hormone replacement therapy (HRT) after RRSO and premature ovarian insufficiency.[11,12] However, none of the current guidelines specifically address BMD and fracture-risk assessment after surgical menopause, as opposed to natural menopause, regardless of age.

The aim of premenopausal oophorectomy is to increase life expectancy and reduce morbidity. Therefore, knowledge on long-term adverse consequences of surgical menopause is important, for both guidance and prevention purposes. Investigating this subject is challenging, since observational studies are prone to bias (e.g. owing to age or HRT use). Randomised trials would offer more reliable results, but would raise ethical concerns. Thus far, no systematic review has assessed the effect of surgical menopause on BMD and fracture prevalence.

Objectives

This review aims to synthesise evidence on the effects of surgical menopause on BMD (g/cm^2), *T*-scores, and fracture prevalence, compared with age-matched control groups without oophorectomy.

Methods

A review protocol was designed based on recommendations of the Centre for Reviews and Dissemination, and is available upon request.[13] Results were reported according to PRISMA and MOOSE guidelines.[14,15]

Eligibility criteria

Studies were eligible for inclusion if they fulfilled the following criteria: original study; in women; control group without oophorectomy; controlling for current age and HRT use; presenting data on bone-related outcomes; English abstracts and full-text available. To ensure that we included studies researching surgical menopause by premenopausal oophorectomy after peak bone mass, we selected studies using 'surgical menopause' or 'premenopausal oophorectomy' to describe the intervention group, and/or in which ≥75% of oophorectomised women were estimated to be aged between 30 and 52 at oophorectomy. Studies not providing age at oophorectomy were excluded, and so were studies on women with oophorectomy for specific conditions (e.g. transsexuality) or for malignancies, BMD level as eligibility criterion, BMD not measured by dual energy X-ray absorptiometry (DXA) at lumbar spine (LS), femoral neck (FN) or hip, case reports and series, and meeting abstracts not published as full texts. Studies were included in meta-analyses when they provided mean BMD in g/cm^2 or *T*-scores and SD or risk estimates for fractures and CIs. Publication date was not restricted.

Information sources and search strategy

Studies were identified by electronic searches in PubMed, EMBASE and Cochrane database with no restriction for publication date (last search: March 1st, 2016) and manual searches of reference lists of included articles. The search strategy was developed with a professional librarian, combining terms related to oophorectomy and bone outcomes, restricting for studies in humans (Appendix S1).

Study selection

Potentially relevant papers were independently screened by two reviewers (IEF and EMA or NT). Studies were excluded when titles and abstracts clearly did not meet the eligibility criteria, and the remaining studies were evaluated for full text. Discrepancies were resolved by dialogue and, if necessary, by appeal to a third reviewer (GHB).

Data extraction

Data were extracted using data extraction forms developed for this review, based on the Cochrane Template (Appendix S2).[16] Data on study design, study population (inclusion and exclusion criteria, sample size), exposure (definition of, and indication for, surgical menopause), outcome measures (BMD/*T*-scores/fractures, measurement device, bones assessed) and methods applied to account for bias were extracted independently by two reviewers (IEF and EMA or NT). Discrepancies were resolved by dialogue and if necessary by appeal to GHB.

Risk of bias for individual studies

Study quality was assessed with an adapted version of the Downs and Black checklist (Appendix S2).[17,18] The following adaptations were made: question 8 was removed; question 14 was replaced by: 'Were patient groups similar with respect to baseline criteria (e.g. current age and/or age at menopause and/or time interval after menopause, body mass index)?'; and question 19 was replaced by: 'Was the intervention (premenopausal oophorectomy) reliably established?'. The maximum score on the adapted checklist was 31 points.

Statistical analysis and data synthesis

Studies were classified according to design (cross-sectional/longitudinal), and categorised according to outcomes [BMD (LS, FN or hip) in g/cm^2, *T*-scores, fracture prevalence]. When studies presented stratified patient characteristics or results, summary estimates were obtained for the total surgical menopause and control groups by pooling the information of subgroups. For studies presenting results for control groups with and without hysterectomy, we only considered those without hysterectomy. For studies providing stratified results, only data controlled for current age were presented. For studies providing results stratified according to age at oophorectomy, only results for age categories between 30 and 52 were included. We preferably used HRT-adjusted estimates, but when not available, data for never users were reported. To explore the impact of age and time interval after menopause on bone, available stratified results were presented in a supplementary table. Additionally, BMD presented by individual studies was plotted against these factors. For cross-sectional studies, mean time interval after menopause was used, while for longitudinal studies, all measurements were plotted.

For meta-analyses, pooled mean differences and SDs (for studies providing BMD in g/cm^2 or *T*-scores for LS or FN/hip) and hazard ratios (HR) with 95% CIs (for studies on fractures) were calculated with Review Manager Software version 5.3 (2014, The Nordic Cochrane Centre, The Cochrane Collaboration, Copenhagen, Denmark). For longitudinal studies we included BMD at the longest follow-up. Random effects models were used to account for variability between studies.

Risk of bias across studies

Statistical heterogeneity was assessed using I^2 and Tau^2 statistics. To explore possible sources of heterogeneity, three sensitivity analyses were performed in order to account for menopausal status of control groups (premenopausal or natural menopause), study quality (Downs and Black score below or above median score), and differences in age at, or time interval after, menopause (similar or not). Publication bias was assessed by visual evaluation of funnel plots, but was not tested because of the small number of studies and their heterogeneity.

Results

The initial search yielded 4888 citations (PubMed, n = 2303, EMBASE n = 2585, Cochrane n = 0) including 1075 duplicates (Figure 1). Title and abstract evaluation resulted in the exclusion of 3748 studies. After full-text evaluation of 65 studies, 16 studies were included.[19–34] Reviewing reference lists of included studies yielded one additional study.[35] Eleven studies provided sufficient data for meta-analysis,[19,20,22–25,28,30–32,35] including two studies by Hadjidakis et al.[23,25] and two by Chittacharoen et al.[19,22] After personal communication with Dr Hadjidakis, one study[23] was excluded from meta-analyses owing to overlapping study populations. We were unable to contact Dr Chittacharoen to assess overlap, therefore both studies were included.

Study characteristics

Characteristics of included studies are summarised in Table S1. Ten studies reported BMD,[19–27,35] three *T*-scores,[26,28,35] and six fracture prevalence.[29–34] Ten studies had cross-sectional[19–26,28,35] and seven longitudinal designs.[27,29–34] All studies comprised a total of at least 43 386 women with surgical menopause (range, 18–16 345) and 437 958 control women (range, 20–408 424) with relevant data available. Included studies used heterogeneous inclusion criteria and strategies to reduce confounding.

Risk of bias within studies

Downs and Black scores are summarised in Table S2; the median score was 18 (range, 6–24) out of 31.

Overall impact of surgical menopause on BMD

Results of studies reporting on BMD in g/cm^2 are summarised in Table 1. Two cross-sectional studies compared BMD in women with surgical menopause with that of premenopausal women; both observed lower BMD after surgical menopause.[19,20] Six cross-sectional studies on BMD described control groups with natural menopause.[21–25,35] One study reported lower LS- and FN-BMD, and one higher FN-BMD after surgical menopause.[22,25] A third study reported higher BMD after surgical menopause compared with population-based reference values.[26] One longitudinal study did not report significant results.[27]

Lumbar spine BMD

Overall meta-analysis of six studies on LS-BMD showed lower BMD after surgical menopause compared with control groups, irrespective of menopausal status [mean difference, −0.05 g/cm^2 (95% CI, −0.10 to −0.01 g/cm^2); Figure S1].

In sensitivity analysis according to menopausal status of the control groups, meta-analysis of two studies with premenopausal control groups showed lower BMD in women with surgical menopause [mean difference, −0.15 g/cm^2 (95% CI, −0.19 to −0.11 g/cm^2); Figure 2]. Meta-analysis of four studies with control groups with natural menopause did not show a significant difference [mean

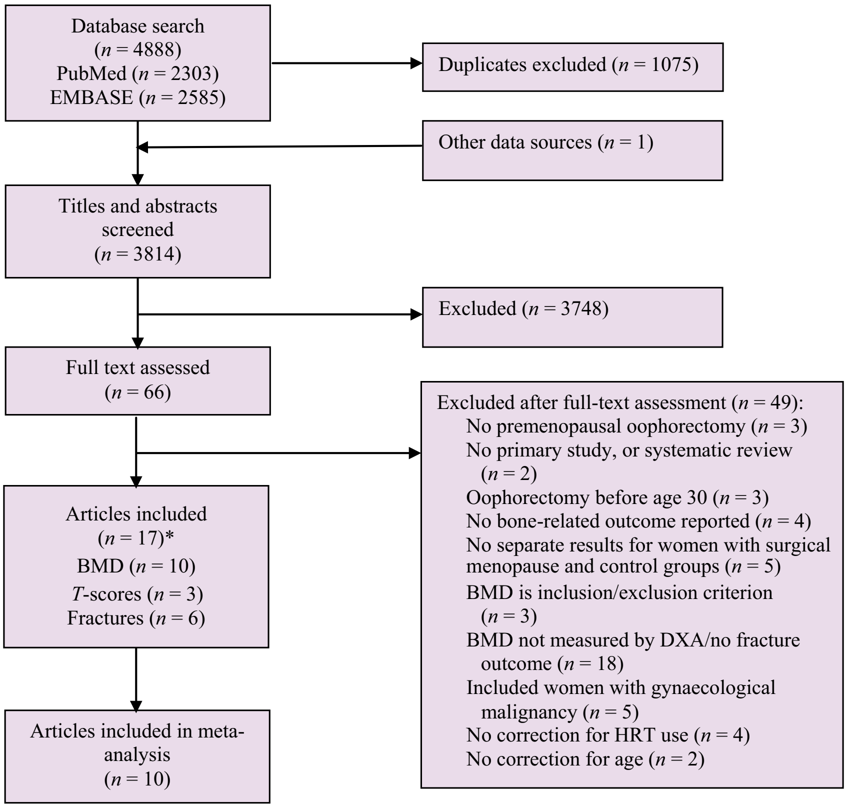

Figure 1. Flow-chart of inclusion process. *Two studies presented both BMD and *T*-scores.

difference, −0.02 g/cm^2 (95% CI, −0.04 to 0.00 g/cm^2); Figure 2].

Both studies with premenopausal control groups had quality scores above the median (Figure S2). In studies with control groups with natural menopause, sensitivity analysis showed lower LS-BMD after surgical menopause in two studies with quality scores below the median and no significant difference in two studies with quality scores above the median (Figure S2).

Sensitivity analysis showed no significant differences in BMD in three studies including women of similar age at, or time since, surgical or natural menopause, but lower BMD after surgical menopause in two studies with a control group with a higher age at, or shorter time since, natural menopause (Figure S3).

Femoral neck BMD

Overall meta-analysis of three studies on FN-BMD showed no difference between women with surgical menopause and control groups, irrespective of menopausal status [mean difference, −0.03 g/cm^2 (95% CI, −0.17 to 0.11 g/cm^2); Figure S1].

Sensitivity analysis according to menopausal status of the control groups included one study with a premenopausal control group reporting lower BMD after surgical menopause (Figure 2). Sensitivity analysis of two studies with control groups with natural menopause did not yield a significant result (Figure 2).

Sensitivity analysis showed no significant differences in BMD in two studies including women of similar age at, or time since, surgical or natural menopause. One study with a control group with a higher age at, or shorter time since, natural menopause showed higher BMD after surgical menopause (Figure S3).

Overall impact of surgical menopause on BMD *T*-scores

Three studies reported lower *T*-scores after surgical menopause, compared with, respectively, premenopausal women, women with natural menopause, and population-based reference data (Table 2).[26,28,35] Meta-analysis of two studies on LS-*T*-scores showed no significant difference after surgical compared with natural menopause [mean difference, −0.29 (95% CI, −0.61 to −0.03); Figure 2 and Figure S1].

Overall impact of surgical menopause on fracture prevalence

Of six studies reporting fracture prevalence (Table 2),[29–34] one reported increased fracture prevalence after surgical compared with natural menopause.[33]

Table 1. Outcomes of all included studies reporting on bone mineral density (BMD) in g/cm^2

Study ID and country	Study group	Outcome	BMD [mean (SD)]
Cross-sectional studies			
Chittacharoen 1997, Thailand	Surgical menopause	LS	0.98 (0.16)*,**
	Controls: Perimenopausal women		1.15 (0.15)
	Surgical menopause	FN	0.76 (0.13)*,**
	Controls: Perimenopausal women		0.93 (0.15)
Yasui 2007, Japan	Surgical menopause (*n* = 112)	LS	0.97 (0.12)*,**
	Controls: Premenopausal women		1.06 (0.15)
Pansini 1995, Italy	Surgical menopause	LS	0.89 (0.13)
	Controls: Spontaneous menopausal women		0.91 (0.15)
Kritz-Silverstein*** 1996, USA	Hysterectomy with bilateral oophorectomy	LS	0.91 (ND)
	Controls: Hysterectomy with conservation of one or both ovaries		0.97 (ND)
	Hysterectomy with bilateral oophorectomy	Hip	0.78 (ND)
	Controls: Hysterectomy with conservation of one or both ovaries		0.81 (ND)
Chitacharoen 1999, Thailand	Surgical menopause	LS	0.99 (0.16)**
	Controls: NM		1.03 (0.15)
	Surgical menopause	FN	0.77 (0.15)**
	Controls: NM		0.80 (0.11)
Hadjidakis 1999, Greece	Surgical menopause	LS	0.86 (0.14)
	Controls: NM		0.88 (0.17)****
	Surgical menopause	FN	0.76 (0.13)
	Controls: NM		0.78 (0.13)****
Ohta***** 2002, Japan	Oophorectomy	LS	0.93 (0.12)
	Controls: NM		0.96 (0.12)
Hadjidakis 2003, Greece	Surgical menopause	LS	0.67 (0.10)
	Controls: NM		0.67 (0.13)******
	Surgical menopause	FN	0.83 (0.12)**
	Controls: NM		0.73 (0.12)******
Hayirlioglu 2006, Turkey	Surgical menopause	LS	1.04 (ND)**
		BMD (%)	89.69
	Controls: USA age-group BMD controls		ND
	Surgical menopause	FN	0.85 (ND)**
		BMD (%)	91.39
	Controls: USA age-group BMD controls		ND
Longitudinal study			
Kritz-Silverstein*** 2004, USA	Hysterectomy with bilateral oophorectomy at first assessment (1988–91)	LS	0.90 (ND)
	Controls: Intact at first assessment		0.91 (ND)
	Hysterectomy with bilateral oophorectomy at follow-up (1992–95)		0.90 (ND)
	Controls: Intact at follow-up		0.91 (ND)
	Hysterectomy with bilateral oophorectomy at first assessment	FN	0.65 (ND)
	Controls: Intact at first assessment		0.65 (ND)
	Hysterectomy with bilateral oophorectomy at follow-up		0.64 (ND)
	Controls: Intact at follow-up		0.64 (ND)
	Hysterectomy with bilateral oophorectomy at first assessment	Hip	0.80 (ND)
	Controls: Intact at first assessment		0.79 (ND)
	Hysterectomy with bilateral oophorectomy at follow-up		0.79 (ND)
	Controls: Intact at follow-up		0.77 (ND)

For pooled estimates, *n* is provided (if available), as it differs from *n* in Table S1.
FN, femoral neck; LS, lumbar spine; ND, no data; NM, natural menopause.
BMD given as g/cm^2 except where indicated.
*Pooled estimates, original results stratified by years since menopause.
**Report significant results.
***Adjusted for multiple factors, most commonly age, body mass index, age at menopause or years since menopause, and hormone replacement therapy (for details see Table S1).
****Pooled estimates, original results stratified by matching criteria (age, years since menopause and both age and years since menopause).
*****Estimates for women with oophorectomy or NM <3 years before enrolment, information for women with event > 3 years before enrolment was not extractable.
******Pooled estimates, original results stratified by age group and type of NM (normal and premature/early).

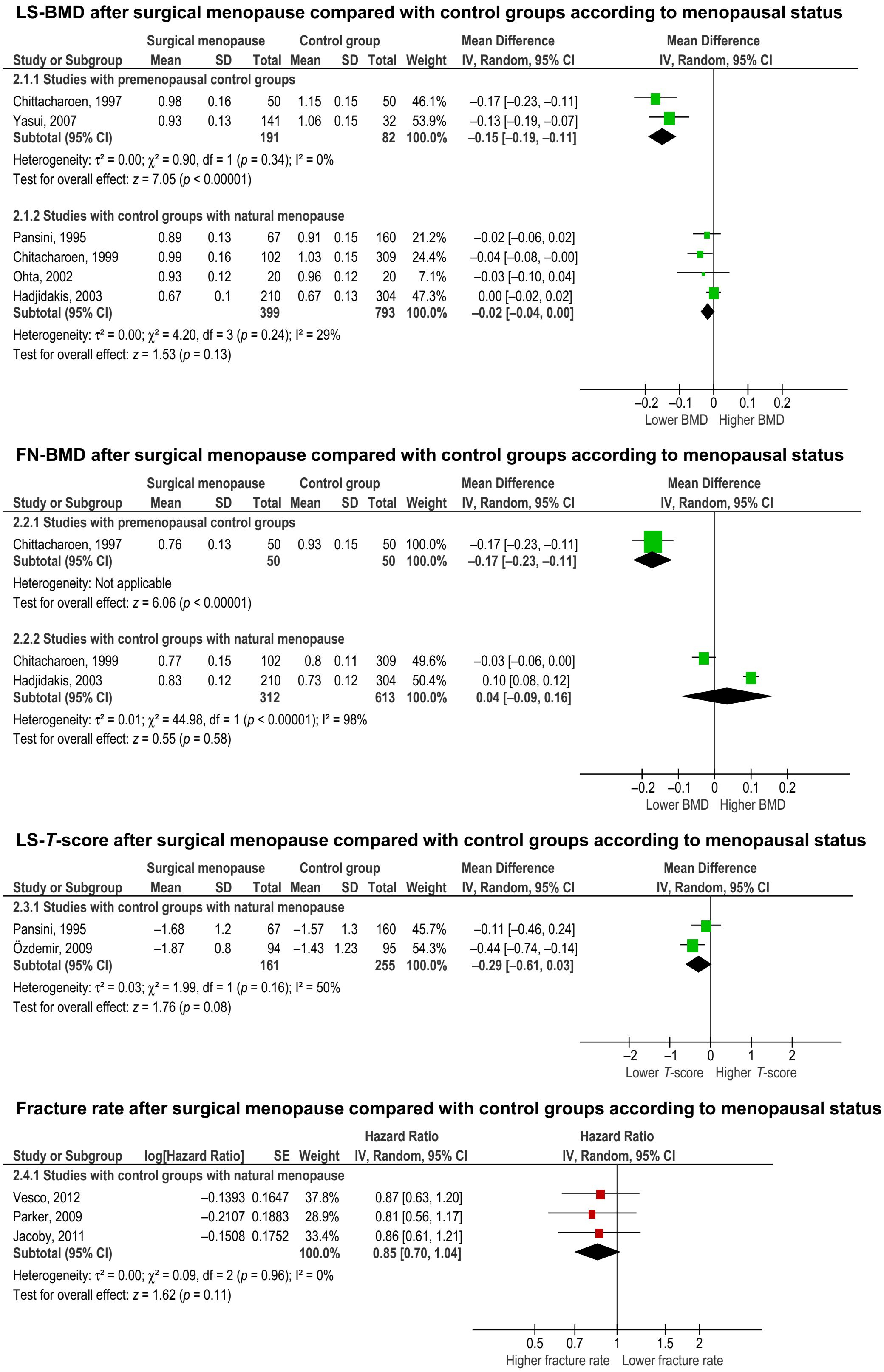

Figure 2. Sensitivity meta-analysis of effect of surgical menopause on BMD, *T*-scores, and fracture rate according to menopausal status of control group.

Three studies provided HRs, one relative risks, one standardised morbidity ratios and one fracture prevalence. Of three studies reporting HRs, two provided information on hip fracture rate only.[30–32] Therefore, we performed a meta-analysis of HR for hip fracture. This meta-analysis revealed no significant difference in hip fracture rate after surgical menopause and after natural menopause [HR, 0.85 (95% CI, 0.70 to 1.04); Figure S1 and Figure 2].

Sensitivity analysis according to quality scores showed no significant differences, either in two studies with a quality score below, or in one study with a quality score above, the median (Figure S2). Sensitivity analysis for differences in age at, and time since, menopause were only possible for the study by Vesco et al.,[30] because the other studies had control groups with hysterectomy before natural menopause.[31,32]

Studies with stratified results

One study presented hip fracture rates stratified by age at hysterectomy and did not observe a significant difference regardless of age at surgery (Table S3).[32]

Another study presented BMD stratified for age and age at natural menopause. LS-BMD was lower after surgical compared with natural menopause after age 45 in women aged 45–50, but not in older women. FN-BMD was never significantly lower after surgical compared with after natural menopause.[25]

Five studies stratified results for time interval after menopause. In two studies with premenopausal control groups, no significant differences were observed for shorter time intervals after surgical menopause (≤3 years and up to 6 months, respectively), but in women with longer time intervals after surgical menopause mean BMD was significantly lower.[19,20] Pansini et al.[35] described no significant differences in BMD or *T*-scores after surgical compared with natural menopause, regardless of time interval after menopause. Hadjidakis et al.[23] found lower LS-BMD after surgical menopause compared with natural menopause in women matched for age and both age and time interval after menopause. FN-BMD was lower after surgical compared with natural menopause in women matched for age, but not in women matched for age and time interval after menopause. Ozdemir et al.[28] reported lower *T*-scores after surgical compared with natural menopause ≤5 years after menopause, but not >5 years after menopause.

Relationship between age and time interval after menopause and BMD

An overall decrease in LS- and FN-BMD was observed with increasing age and time interval after menopause (Figure S4). By subjective visual evaluation, no evident differences between surgical and natural menopause were observed.

Risk of bias across studies

Visual evaluation of the funnel plots did not show any evidence of publication bias (Figure S5).

Discussion

Main findings

Within this systematic review and meta-analysis, LS- and FN-BMD were lower in women with surgical menopause than in premenopausal age-matched women. When comparing age-matched women with surgical with those with natural menopause, no consistent differences were observed for either BMD or fracture rate.

Strengths and limitations of the study

This is the first systematic review and meta-analysis addressing bone health after surgical menopause. Only studies reporting on LS- and FN/hip-BMD measured by DXA were included, which is the parameter advised by international guidelines.[7–10,36–38] The advantage of limiting our analyses to studies carried out with DXA are its clinical applicability, reliability, and a reduction in heterogeneity. Studies using other measures to assess bone health were excluded and information on these assessments in included papers was not analysed. Still, these assessments might provide relevant information on bone health after surgical menopause.

The quality of this meta-analysis is hampered by the quality of the included studies. Quality scores were particularly low on external validity and internal validity/confounding, indicating that included studies are prone to bias. Furthermore, heterogeneity in meta-analysed studies on BMD and *T*-scores was high. We considered it appropriate to meta-analyse their results, because all included studies aimed to investigate the same effect, and meta-analysis will provide information on the magnitude and sources of heterogeneity.[39] Heterogeneity was taken into account by stratification into several subgroup analyses. Other possible strategies for overcoming heterogeneity would be meta-regression analysis or individual patient data meta-analysis. However, owing to the small number of studies and the heterogeneity of the populations under study, this was not an option. For the same reason, objective assessment of publication bias across studies by testing funnel-plot asymmetry was not possible.[16]

Interpretation

Studies on BMD with a premenopausal control group

BMD was lower in women after surgical menopause than in age-matched premenopausal women. This difference tended to be larger with a longer time interval after surgical

Table 2. Outcomes of all included studies reporting on *T*-scores and fracture rate

Study ID and country	Study group	Outcome	Result
Studies on *T*-scores in mean (SD)			
Cross-sectional			
Pansini 1995, Italy	Surgical menopause	*T*-score LS	–1.68 (1.20)
	Controls: Spontaneous menopausal women		–1.57 (1.30)
Ozdemir 2009, Turkey	Surgical menopause	*T*-score LS	–1.87 (0.80)
	Controls: NM		–1.43 (1.23)
	Surgical menopause	*T*-score Hip	–1.30 (0.87)*
	Controls: NM		–0.87 (0.98)
Hayirlioglu 2006, Turkey	Surgical menopause	*T*-score LS	–0.96 (ND)*
		T-score FN	–0.64 (ND)*
	Controls: USA age-group BMD controls		ND
Studies on fracture prevalence (95% CI)			
Longitudinal studies			
Banks 2009, UK	Surgical menopause	RR hip fracture	1.20 (0.94–1.55)**
	Controls: NM		1
Vesco*** 2012, USA	Surgical menopause (n = 433)	HR hip fracture	0.87 (0.63–1.21)**
	Controls: NM (n = 3683)		1
	Surgical menopause (n = 433)	HR wrist fracture	1.08 (0.76–1.54)**
	Controls: NM (n = 3683)		1
	Surgical menopause (n = 433)	HR non-vertebral fracture	1.10 (0.92–1.31)**
	Controls: NM (n = 3683)		1
Parker**** 2009, USA	Hysterectomy with bilateral oophorectomy	HR hip fracture	0.81 (0.56–1.17)**
	Controls: Hysterectomy with ovarian conservation		1
Jacoby****,***** 2011, USA	Hysterectomy with bilateral oophorectomy	HR hip fracture	0.86 (0.61–1.23)**
	Controls: Hysterectomy with ovarian conservation		1
Johansson 1993, Sweden	Hysterectomy with bilateral oophorectomy	Prevalence radius/humerus/ vertebral/ hip/tibial condyle fractures	38.9%*
	Controls: Neither hysterectomy nor bilateral oophorectomy		21.0%
Melton III****,****** 1996, USA	*Surgical menopause*		
	age at ED 35–44	SMR vertebral fracture	2.3 (0.8–4.9)
	age at ED 45–49		1.8 (0.9–3.4)
	Controls: General population of Rochester		1
	Surgical menopause		
	Age at ED 35–44	SMR hip fracture	1.1 (0.1–4.0)
	Age at ED 45–49		0.9 (0.2–2.3)
	Controls: General population of Rochester		1
	Surgical menopause		
	Age at ED 35–44	SMR forearm fracture	2.0 (0.9–3.6)
	Age at ED 45–49		1.3 (0.7–2.3)
	Controls: General population of Rochester		1

For pooled estimates, n is provided (if available), as it differs from n in Table S1.
ED, estrogen deficiency; FN, femoral neck; HR, hazard ratio; LS, lumbar spine; NM, natural menopause; RR, relative risk; SMR, standardised morbidity ratio.
*Report significant results.
**Adjusted for multiple factors, most commonly age, body mass index, age at menopause or years since menopause and hormone replacement therapy (for details see Table S1).
***Including only results for never HRT users.
****Including only results for women with surgery at premenopausal age (< 52; n for subgroups not stated in the original study).
*****Pooled estimates, results originally presented stratified by age at menopause.
******Results for subgroup with ED < 35 years of age are not presented (because it includes women with oophorectomy before peak bone mass).

menopause.[19,20] Studies measuring BMD shortly after surgical menopause are unlikely to find significant differences. In our meta-analysis of two studies with premenopausal control groups on the longest available follow-up, mean differences were approximately one SD (LS-BMD, -0.15 g/cm^2; FN-BMD, -0.17 g/cm^2).[19,20] A difference of one SD in BMD was associated with a two-fold increase in age-specific fracture risk.[40,41] When interpreting this relative increase, baseline fracture risks, which are relatively low for women aged <50 years, need to be considered.[42]

Studies on BMD with control group with natural menopause

The reported directions and magnitudes of differences in BMD between women with surgical and natural menopause were inconsistent. Two studies reported lower BMD after surgical menopause[22,26] and one reported lower BMD after natural menopause.[25] Other studies did not report significant overall differences.

Studies with a control group with natural menopause had diverse inclusion and exclusion criteria (Table S1), which was reflected by high heterogeneity in the meta-analyses.

Age at and time interval after menopause in control groups varied widely among studies, and some did not provide information on it.[21,22,24–27] Age at natural menopause is expected to be higher and time interval after natural menopause should be shorter than after surgical menopause, because appropriate control women should not be postmenopausal before women with surgical menopause. Women with natural menopause experience a perimenopausal period with bone loss before menopause.[43] The effect of surgical menopause might be underestimated in studies with a control group with lower or similar age at, or shorter or similar time interval after, natural menopause.[21–23,35] This hypothesis is supported by the findings from the sensitivity analyses according to age at, and time since, menopause on LS-BMD.

In addition, of the studies including a control group with higher age at, and shorter time interval after, natural menopause, studies in younger women showed lower BMD after surgical menopause compared with natural menopause, but studies in older women found similar or higher BMD after surgical menopause.[25,27] This might indicate that an effect of surgical menopause on BMD becomes less relevant in older age.

Only studies that accounted for the use of HRT were included. Several studies excluded women using HRT, which may have caused selection bias, because HRT was likely to be prescribed to women at high risk for bone loss or having low BMD.[22–26,35] This may have caused overestimation of BMD in the surgical menopause group.

All studies retrospectively assessed oophorectomy status at inclusion and prospectively measured BMD. Most studies excluded women using medication affecting bone.[23–26,35] Since an effect of surgical menopause on bone is expected, physicians might be more concerned about bone loss and act preventatively. This may have induced selection bias, as women at high risk for bone loss may have been excluded because they were using medication affecting bone at inclusion.

Only studies measuring BMD by DXA were included, however, different DXA machines were used (Hologic, Lunar and Norland). BMD values may vary significantly when measured by different devices,[44] which could be a source of heterogeneity. Within studies all women were measured with the same device, so within studies this was not a source of bias.

Studies on *T*-scores after surgical menopause had comparable results to studies on BMD in g/cm^2, and were vulnerable to the same types of bias.

Studies on fracture prevalence after surgical menopause

One of six studies reported higher fracture prevalence in women with surgical menopause.[29–34] However, age varied widely between studies and some did not provide data on possible confounding factors (e.g. age, age at menopause, time interval after menopause, and time of fracture). Furthermore, menopausal status was assessed retrospectively, which may have caused selection and recall bias. Additionally, when including women aged 65–70, survival bias is likely.[30,33] Fractures (especially hip/femoral) increase mortality in the elderly.[45] If surgical menopause increases fracture prevalence, it is possible that women with fractures due to surgical menopause have worse survival.

Within studies researching women with hysterectomy with and without oophorectomy,[31,32] hysterectomy might affect age at menopause, thus BMD in a control group with hysterectomy might be lower than in a control group without hysterectomy.[46]

Conclusions

These results suggest that directly after surgical menopause BMD decreases significantly, since women with surgical menopause have lower BMD than premenopausal age-matched controls. However, after the age of natural menopause, BMD and fracture rates seem to be comparable for women with natural and surgical menopause.

The results summarised in this systematic review and meta-analysis should be interpreted with caution, as included studies were prone to bias. Currently, the available literature does not provide sufficient support for recommending systematic BMD screening solely because of surgical menopause. This review might guide further studies in this field to support optimal clinical management.

Suggestions for further research

To obtain reliable estimates of the effect of surgical menopause on bone, prospective inclusion of women with surgical menopause is preferred. An age-matched control group would allow correction for the effect of age on bone. As surgical menopause is, by definition, carried out before the age of natural menopause, follow-up in the control group starts at premenopausal age. Groups should not be selected on fracture history or use of medication affecting the bone. These factors can be considered in sensitivity analyses, as they may influence choice for oophorectomy. A time-to-event approach (e.g. time to fracture or to osteoporosis), or repeated measurements of BMD, with a standardised DXA procedure, could be adopted.

Disclose of interests

None declared. Completed disclosure of interests form available to view online as supporting information.

Contribution to authorship

IEF, NT, EMA, RHJAS, MJEM and GHB were involved in the conception and planning of the review protocol. IEF, NT and EMA executed the literature search, data extraction and quality assessment. IEF and NT performed the meta-analysis and drafted the manuscript. IEF, NT, EMA, RHJAS, MJEM and GHB revised and approved the final manuscript.

Details of ethics approval

Ethics approval was not required, as this is a systematic review of published studies.

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Supporting Information

Additional Supporting Information may be found in the online version of this article:

Appendix S1. Search strategy for PubMed and EMBASE.

Appendix S2. Data collection form, including adapted Downs and Black Checklist.

Figure S1. Overall meta-analysis of the effect of surgical menopause on BMD, *T*-scores and fracture rate.

Figure S2. Sensitivity meta-analysis of the effect of surgical menopause on BMD, *T*-scores and fracture rate according to quality score assessed by Downs and Black checklist (below or above the median score of 18 out of 31 possible points).

Figure S3. Sensitivity meta-analysis of the effect of surgical menopause on BMD, *T*-scores and fracture rate according to differences in age at or time interval after surgical or natural menopause.

Figure S4. Relationship between age and time interval after menopause and BMD.

Figure S5. Funnel plots for the overall meta-analyses on BMD, *T*-scores and fracture rate after surgical menopause.

Table S1. Characteristics of included studies.

Table S2. Assessment of bias in studies (based on Downs and Black checklist).

Table S3. Outcomes of studies with stratified results. ■

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